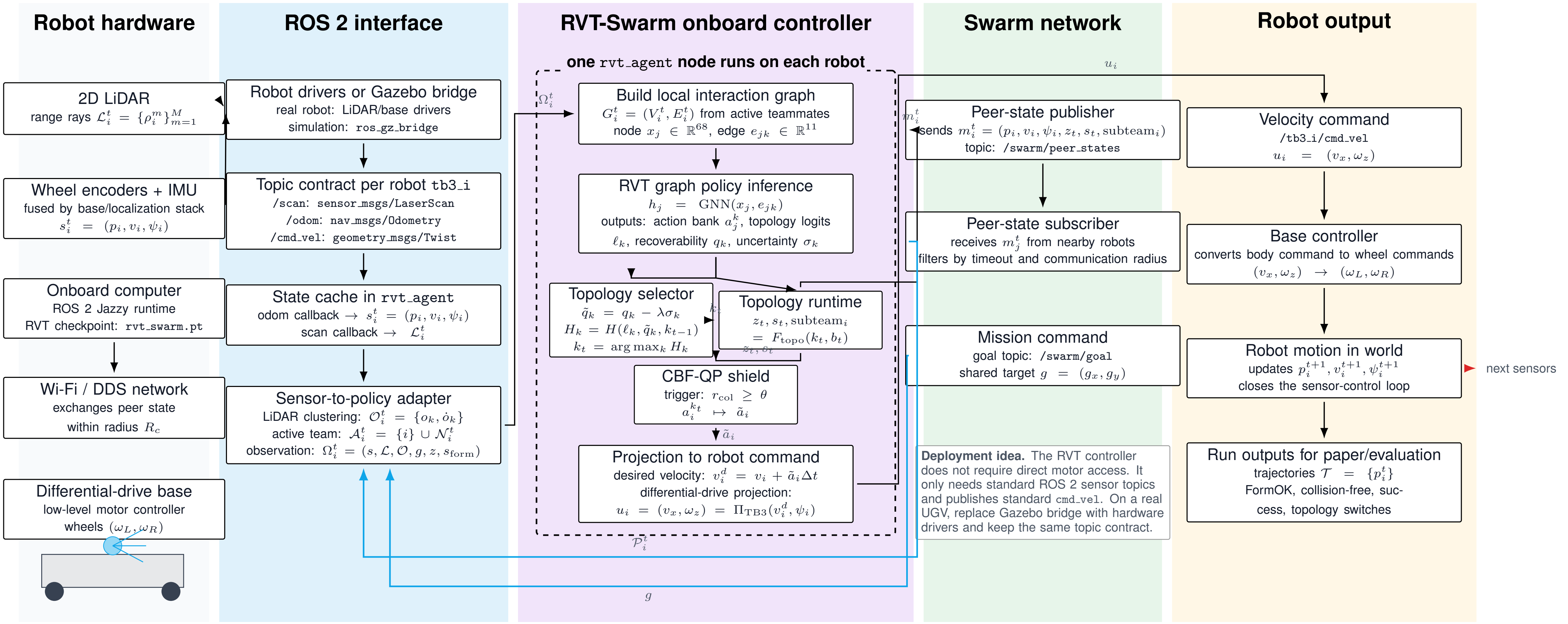

Robot hardware
ROS 2 interface
RVT-Swarm onboard controller
Swarm network
Robot output
one rvt_agent node runs on each robot
2D LiDAR
range rays $\mathcal{L}_i^t = \{\rho_i^m\}_{m=1}^M$
Wheel encoders + IMU
fused by base/localization stack
$s_i^t = (p_i, v_i, \psi_i)$
Onboard computer
ROS 2 Jazzy runtime
RVT checkpoint: rvt_swarm.pt
Wi-Fi / DDS network
exchanges peer state
within radius R_c
Differential-drive base
low-level motor controller
wheels (ω_L, ω_R)
Robot drivers or Gazebo bridge
real robot: LiDAR/base drivers
simulation: ros_gz_bridge
Topic contract per robot tb3_i
/scan: sensor_msgs/LaserScan
/odom: nav_msgs/Odometry
/cmd_vel: geometry_msgs/Twist
State cache in rvt_agent
odom callback → $s_i^t = (p_i, v_i, \psi_i)$
scan callback → $\mathcal{L}_i^t$
Sensor-to-policy adapter
LiDAR clustering: $\mathcal{O}_i^t = \{o_k, \dot{o}_k\}$
active team: $\mathcal{A}_i^t = \{i\} \cup \mathcal{N}_i^t$
observation: $\Omega_i^t = (s, \mathcal{L}, \mathcal{O}, g, z, s_{\text{form}})$
Ω_i^t
Build local interaction graph
$G_i^t = (V_i^t, E_i^t)$ from active teammates
node $x_j \in \mathbb{R}^{68}$, edge $e_{jk} \in \mathbb{R}^{11}$
RVT graph policy inference
$h_j = \text{GNN}(x_j, e_{jk})$
outputs: action bank a_j^k, topology logits
ℓ_k, recoverability q_k, uncertainty σ_k
Topology selector
$\tilde{q}_k = q_k - \lambda\sigma_k$
$H_k = H(\ell_k, \tilde{q}_k, k_{t-1})$
$k_t = \arg\max_k H_k$
k_t
Topology runtime
$z_t, s_t, \text{subteam}_i$
$= F_{\text{topo}}(k_t, b_t)$
z_t, s_t
CBF-QP shield
trigger: $r_{\text{col}} \geq \theta$
$a_i^{k_t} \mapsto \tilde{a}_i$
$\tilde{a}_i$
Projection to robot command
desired velocity: $v_i^d = v_i + \tilde{a}_i \Delta t$
differential-drive projection:
$u_i = (v_x, \omega_z) = \Pi_{\text{TB3}}(v_i^d, \psi_i)$
$\mathcal{P}_i^t$
g
u_i
m_i^t
Peer-state publisher
sends $m_i^t = (p_i, v_i, \psi_i, z_t, s_t, \text{subteam}_i)$
topic: /swarm/peer_states
Peer-state subscriber
receives m_j^t from nearby robots
filters by timeout and communication radius
Mission command
goal topic: /swarm/goal
shared target $g = (g_x, g_y)$
Deployment idea. The RVT controller does not require direct motor access. It only needs standard ROS 2 sensor topics and publishes standard cmd_vel. On a real UGV, replace Gazebo bridge with hardware drivers and keep the same topic contract.
Velocity command
/tb3_i/cmd_vel
$u_i = (v_x, \omega_z)$
Base controller
converts body command to wheel commands
$(v_x, \omega_z) \rightarrow (\omega_L, \omega_R)$
Robot motion in world
updates $p_i^{t+1}, v_i^{t+1}, \psi_i^{t+1}$
closes the sensor-control loop
next sensors
Run outputs for paper/evaluation
trajectories $\mathcal{T} = \{p_i^t\}$
FormOK, collision-free, success, topology switches